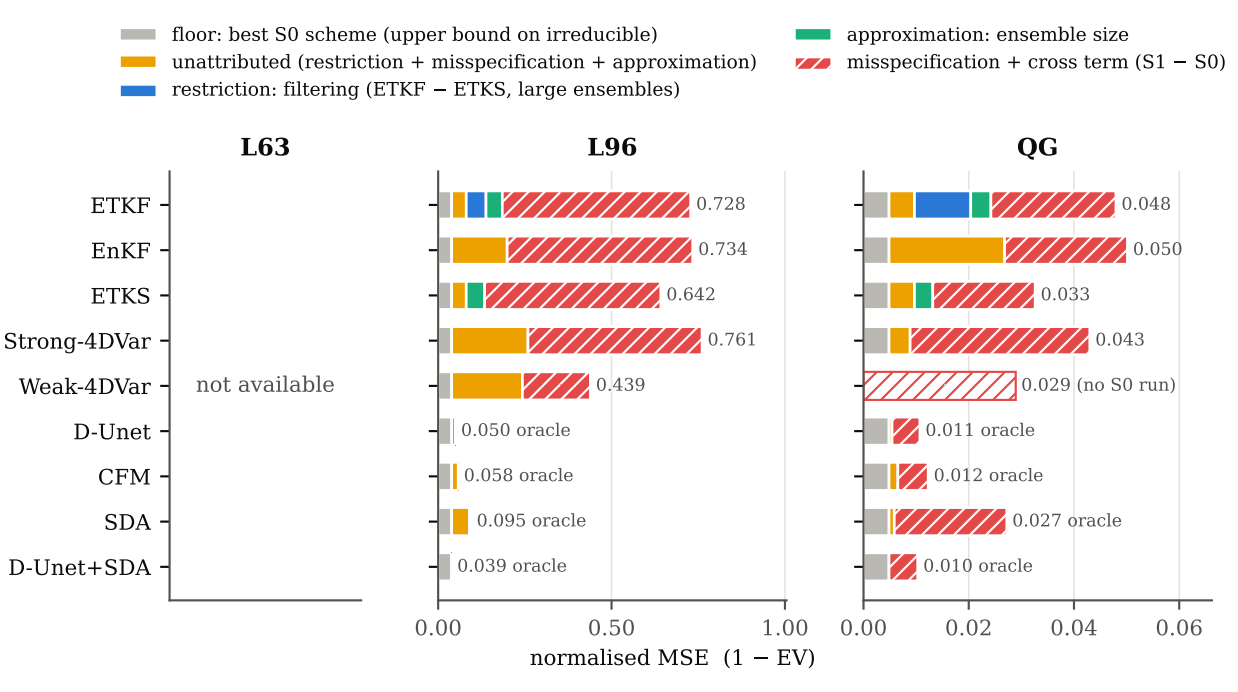
floor: best S0 scheme (upper bound on irreducible)
unattributed (restriction + misspecification + approximation)
restriction: filtering (ETKF − ETKS, large ensembles)
approximation: ensemble size
misspecification + cross term (S1 − S0)
L63
L96
QG
ETKF
EnKF
ETKS
Strong-4DVar
Weak-4DVar
D-Unet
CFM
SDA
D-Unet+SDA
not available
0.728
0.734
0.642
0.761
0.439
0.050 oracle
0.058 oracle
0.095 oracle
0.039 oracle
0.048
0.050
0.033
0.043
0.029 (no S0 run)
0.011 oracle
0.012 oracle
0.027 oracle
0.010 oracle
0.00
0.50
1.00
0.00
0.02
0.04
0.06
normalised MSE (1 − EV)